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Role: AI/ML Intern

DAY – 4

TASK-1. Pose Detection Accuracy Improvements

Research Study

Pose detection is a computer vision technique used to identify key body points such as shoulders, elbows, knees, and hips. Accurate pose detection is important for fitness tracking, motion analysis, gaming, and virtual try-on systems.

Why Pose Detection Accuracy Needs Improvement

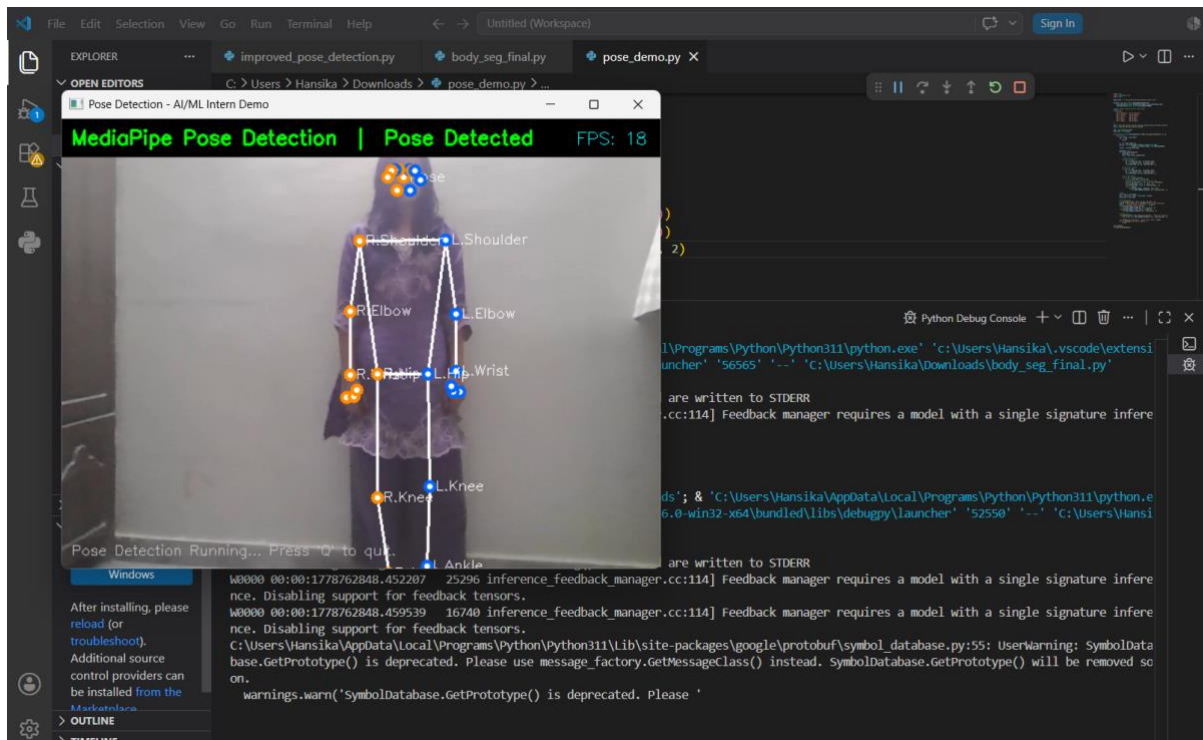
Pose detection accuracy is important because small errors in body tracking can affect the entire AI system. In virtual try-on and body measurement applications, incorrect landmark detection may place clothes incorrectly or produce wrong measurements.

Improvements Tested

- Better lighting conditions improved landmark visibility.
- Higher webcam resolution increased detection accuracy.
- Reducing background clutter helped the model focus on the human body.
- Maintaining proper distance from the camera improved body tracking.

Output Observation

Today's pose detection output is significantly more accurate compared to yesterday. The improved accuracy is mainly due to better lighting conditions during testing. With proper front-facing light, the model was able to detect all 33 landmarks clearly with high visibility scores above 0.9 for key joints like shoulders and hips.



TASK-2. Research on AI Virtual Fitting System

Research Study

Virtual try-on systems allow users to digitally test clothes using AI. These systems combine pose estimation, body segmentation, and cloth simulation technologies.

Working Process

1. Detect body landmarks.
2. Separate the body from background.
3. Resize and align clothing image.
4. Overlay clothing on the u

Working Flow of Virtual Try-On AI Model

1. Body Detection

The AI first detects the person's body and identifies important body landmarks such as shoulders, arms, waist, and legs.

2. Body Segmentation

The system separates the human body from the background to clearly identify body boundaries.

3. Clothing Alignment

The selected clothing image is resized and adjusted according to the user's body shape and posture.

4. Virtual Overlay

The AI places the clothing digitally onto the user's body image in real time.

5. Final Output Generation

The system generates the final virtual try-on image showing how the outfit looks on the user.

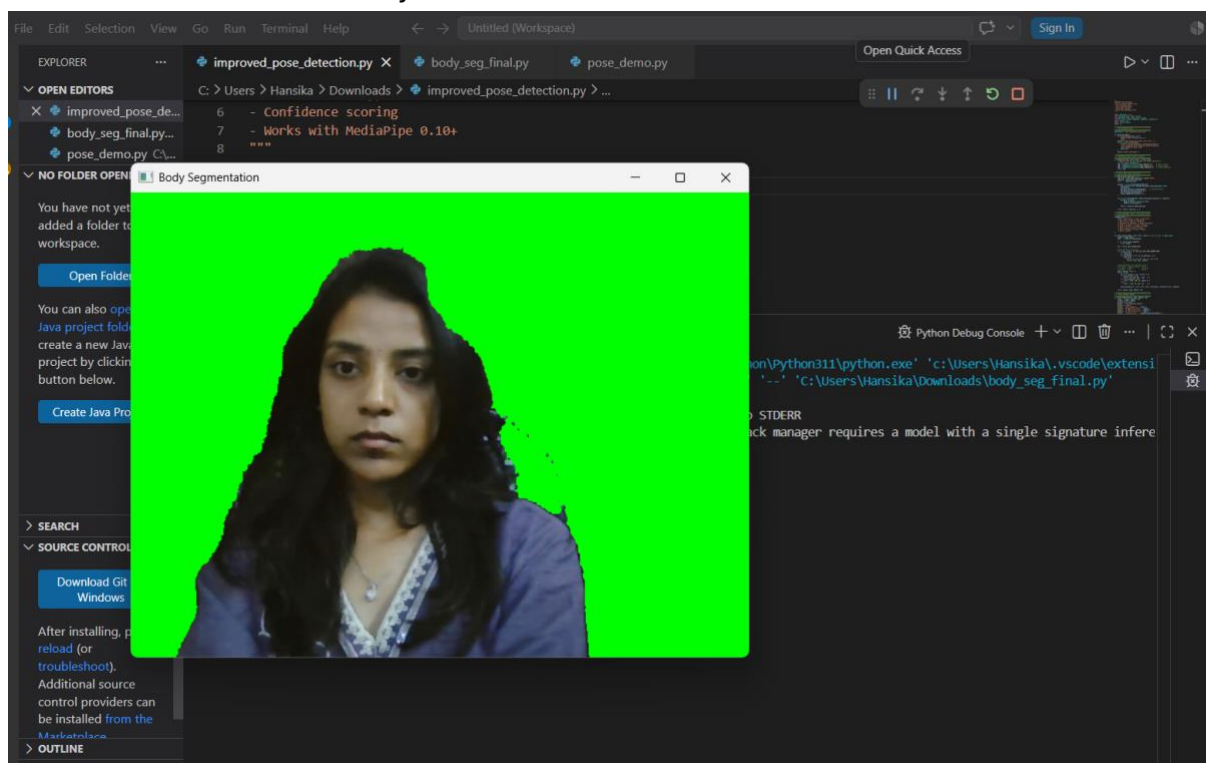
TASK-3. Testing Body Segmentation Methods

To test different AI body segmentation methods and understand how accurately they separate the human body from the background.

What I Did

I tested body segmentation models using webcam input in Python. I used MediaPipe segmentation to identify the human body and remove the background in real time. I also observed how segmentation changes under different lighting conditions and camera positions.

worked smoothly even with movement



Outcome From My Testing

- The model was able to separate the body from the background successfully.
- Good lighting improved segmentation quality.
- Background noise slightly affected edge detection.
- Real-time segmentation worked smoothly with webcam input.
- The body outline became more accurate when the user stood closer to the camera.

My Observations

- Good lighting improves segmentation accuracy.
- Background noise affects body edge detection.
- Real-time segmentation works smoothly with webcam.
- Closer camera distance gives better body outline detection.
- Media Pipe segmentation is fast and lightweight.
- Body separation becomes difficult during fast movement.

TASK-4. Clothing Recommendation Logic

To create an AI-based clothing recommendation system that suggests outfits according to user preferences and conditions.

What I Did The system takes user inputs such as:

- Body Type
- Occasion
- Weather
- Style Preference

Based on these inputs, the recommendation logic selects suitable:

- Outfit type
- Fabric type
- Fit style
- Colour combinations

Code:

```
improved_pose_detection.py  body_seg_final.py  pose_demo.py  recomendation.py X
C: > Users > Hansika > Downloads > recomendation.py > get_recommendations
1  # AI Clothing Recommendation System
2
3  def get_recommendations(body_type, occasion, weather, preferred_style, preferred_color):
4
5      recommendations = []
6
7      # Body type based fit
8      if body_type == "slim":
9          fit = "Slim Fit"
10     elif body_type == "athletic":
11         fit = "Regular Fit"
12     elif body_type == "plus":
13         fit = "Relaxed Fit"
14     else:
15         fit = "Regular Fit"
16
17     # Occasion + Style Recommendation
18     if occasion == "formal":
19
20         if preferred_style == "western":
21             outfit = "Blazer + Formal Shirt + Trousers"
22
23         elif preferred_style == "ethnic":
24             outfit = "Kurta + Churidar"
25
26         else:
27             outfit = "Formal Shirt + Trousers"
28
29     elif occasion == "casual":
30
31         if preferred_style == "western":
32             outfit = "T-Shirt + Jeans"
33
```

Output

```
PS C:\Users\Hansika\Downloads> & 'C:\Users\Hansika\AppData\Local\Programs\Python\Python311\python.exe' 'c:\Users\Hansika\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '54559' '--' 'C:\Users\Hansika\Downloads\recomendation.py'
=====
AI Clothing Recommendation
=====
Enter Body Type (slim/athletic/plus): athletic
Enter Occasion (formal/casual/party): party
Enter Weather (hot/cold/rainy): hot
Enter Preferred Style (western/ethnic): westren
Enter Preferred Color (bright/dark/neutral): neutral

===== Recommendation =====
Outfit           : Party Wear Shirt + Jeans
Fit              : Regular Fit
Fabric           : Cotton Fabric
Recommended Colors : White / Grey / Beige
-----
```

Why AI is Better Than Manual Search

A normal user searching online has to apply multiple filters manually and still may not find the right fit. AI combines all these conditions simultaneously and gives a single accurate recommendation in seconds. As the user interacts more with the system the AI learns their preferences and becomes more accurate

My Testing Outcome

- The system generated outfit recommendations successfully.
- Different inputs produced different clothing combinations.
- The recommendation logic worked correctly for casual, formal, and party wear.

TASK-5. Media Pipe vs Open Pose Comparison

Media Pipe and Open Pose are popular AI frameworks used for human pose detection. Both systems detect body landmarks such as hands, legs, shoulders, and facial points, but they differ in speed, accuracy, and system requirements.

Comparison Study

Feature	Media Pipe	Open Pose
Speed	Faster	Slower
Accuracy	Good	Very High
System Requirement	Low	High
GPU Requirement	Not necessary	Recommended
Real-Time Performance	Excellent	Moderate
Ease of Installation	Easy	Complex

My Observations

- Media Pipe worked smoothly in real time using a webcam.
- Open Pose provides more detailed body key points.
- Media Pipe is lightweight and beginner friendly.
- Open Pose requires more processing power and setup.